# Automation Laboratory Manual

## Module 1 - Practical 1: Setting Up the Environment

Aim: prepare the Python automation environment for the semester.

Software required: Python 3.11, Google Chrome 120, and Visual Studio Code.

Install the libraries with the command pip install selenium pytest webdriver-manager openpyxl

The driver is created by calling webdriver.Chrome(service=Service(ChromeDriverManager().install())).

Every script must be saved in the folder C:/automation/practicals before execution.

Result: the environment is ready and the sample script runs without errors.

## Module 1 - Practical 2: Login Form Test

Aim: automate the login form of the demo application and verify the page title.

Open the login form with driver.get("https://the-internet.herokuapp.com/login").

Enter the username tomsmith into the field with id user-name.

Enter the password SuperSecretPassword! into the field with id password.

Click the Login button and wait for the flash message to render.

assert "You logged into a secure area!" in driver.page_source

Viva question: why is the flash message asserted instead of the URL?

## Module 1 - Practical 3: Excel Driven Test

Aim: read test data from a workbook and write the verdict back.

Input workbook: students.xlsx with columns name, marks, and grade.

The report is written to results.xlsx in the same folder as the script.

Openpyxl loads the sheet with wb = openpyxl.load_workbook("students.xlsx").

Each row is passed to the test method through pytest.mark.parametrize.

A row fails when marks is below 35, and the reason is stored in the remarks column.

## Module 2 - Practical 4: Wait Handling

Aim: compare implicit, explicit, and hard waits on a slow page.

driver.implicitly_wait(10)

Wait an additional time.sleep(3) before capturing the screenshot.

Set the explicit wait as WebDriverWait(driver, 15).until(EC.element_to_be_clickable((By.ID, "cart"))).

Observation: the implicit value applies to every find element call in the session.

The timeout value is recorded in the lab record as 15 seconds.

## Module 2 - Practical 5: Alerts and Frames

Aim: handle a JavaScript confirm dialog and an inner iframe.

Switch with driver.switch_to.alert then call alert.accept().

Use alert.dismiss() to reject the dialog without confirming.

Move into the frame with driver.switch_to.frame("iframe-result").

## Module 3 - Practical 6: Project Report

Aim: run the regression suite and summarise the outcome.

The suite is executed with pytest -v --html=report.html --self-contained-html.

Twelve test cases are written in total, of which nine are automated and three are manual.

Defect status moves from New to Assigned, Verified, and finally Closed.

The report must be signed by the lab instructor before the internal assessment.

Viva question: what is the difference between severity and priority?

Semester evaluation: 30 marks for practicals, 20 for the record, and 50 for the written paper.